EXUN 2026-27 · HACKATHON · IDEATION

EXUN Case Terminal

Case #1: Wenoti Arcade

Team	Team AFBBS: Medhansh Sharma, Aryaveer Dalal, Mudit Shakya
Event	Exun 2026-27 Hackathon
Deliverables	This Ideation, a website, and a pitch deck
Play it	wattsonbuild.vercel.app
For judges	wattsonbuild.vercel.app/?supervisor=1 opens every folder
Code	github.com/medhu0505/Wattson_Build

THE IDEA IN ONE SENTENCE

We didn't build a website about the case. We built the case: EXUN's own case terminal, where a newly recruited agent works Case #1 with the real case files, meter data and game code, discovers for themselves that a video game character has crossed into the real world, and then has to decide what to do when the agency is suspended.

By the numbers

9	6 + 1	21	6	231	0
case files used, word for word	folders, then an ending	details to pin, 7 of them key	decoding tools	automated tests	servers: a static site

01 The brief, as we read it

The Hackathon handed us a case, not a feature list. Before designing a single screen, we read every exhibit and asked one question of each: what could this become in a player's hands?

Exhibit	What it contains	What it became
Case Debrief	The assignment: Wenoti Arcade on 92nd Avenue, its power surges, and the Machine ID	The briefing, the first evidence, and the lock on folder 03
Transcript A	The 22:31 encounter with POI-1, described as "a cosplayer"	Evidence to pin, and redactions that lift as the case goes on
Asset Integrity Report	A physical inspection that finds nothing conclusive	Details that reward careful reading, some of which turn out not to matter
Energy Usage Dossier	14 days of meter readings and reported usage	The forensics puzzle: a surge and a mismatch
OriginalCode.txt and AlteredCode.txt	Two versions of the game's C++ code	The machine terminal: a real code comparison, and a game world drawn from the code
Progress Assesment Log	Agent Nano's conclusion, and the question WHY?	The confirmation after the reveal, and the hook for Case #2
Suspension Notice	The Government suspends EXUN on 27 October, calling the findings inconclusive	The ending: it arrives once the report is filed, and the agent must acknowledge it
Note from Nano	A secret goodbye: a "digital jailbreak" around the world, and the question "What are you going to do now?"	The final choice: comply, or keep investigating off the record

What we noticed

- The meter readings hide a day that doesn't fit: 2,684 kWh, against a typical 28.1.
- The Machine ID, MJWHKZLTNBUXE5A, isn't random. It is Base32 for the word **blueshirt**.
- The two code files differ by exactly one thing: the Character class is gone.
- The "cosplayer" checks the time on a bare wrist.
- The last two files turn everything around: the official record calls the case inconclusive, while Nano's note says it was only the beginning.

The puzzles were already in the evidence. Our job was to let the player find them.

02 The concept

EXUN Case Terminal is the agency's own desk, opened in a browser. The player signs in as a new recruit on the team of Conis Nikrobo, the EXUN officer the Case Debrief is addressed to, and works the case themselves: reading, pinning, measuring, decoding and comparing. The site is a tool inside the story world, not a page about it.

Four design pillars

Pillar	What it means	How we hold ourselves to it
The player works the case	Every folder asks for what an analyst would do: read, pin, measure, decode, compare	Each folder opens only once a real task is done
Canon is never contradicted	Documents are quoted word for word, typos included, and every number comes from the files	Automated tests check our quotes and figures against the canon text
The reveal is earned	Nothing states the answer before the player finds it	Nano's log stays sealed until the machine is solved; the Suspension Notice isn't even listed until the report is filed
We can explain every line	Plain JavaScript, no framework, and the logic kept apart from the screens	231 tests over the logic, and plain comments throughout

03 Who it's for

Audience	What they need	What we built
The player	A mystery that respects their intelligence	Real evidence, real tools, and hints that point the way without giving it away
The judges	To see everything, quickly	Supervisor mode (?supervisor=1): every folder open, a separate save, and a 3½-minute demo route
Our team	To own and explain every part	Logic in small, tested modules, and all the story text in one file (content/story.json)

04 The player's journey

The case is a folder of six stages. Each stays sealed until the one before it is cleared, so the story is paced by discovery, not by a timer. Filing the report brings the ending.

Folder	What the player does	Story beat	Opens the next step
00 Intake	Picks a codename, agrees to the oath, gets an ID card stamped Admitted, reads the briefing	Recruited to EXUN	Accepting the assignment
01 Case Files	Reads the Debrief, Transcript A and the Asset Integrity Report; pins details to a corkboard (21 to find over the case, 7 of them key)	The arcade, the machine, the stranger	All 7 key details pinned, then filed
02 Energy Forensics	Runs an analysis of the meter data, reads the chart, files a Discrepancy Referral	2,684 kWh on 23 October	A correct referral: the surge and the mismatch
03 Signal Decoder	Decodes the Machine ID with six tools or auto-detect	The ID is a key: blueshirt	Entering the decoded key
04 Machine Terminal	Uses the arcade cabinet's own screen: LOGS, DIFF and RENDER	The character is missing from the game	All three checks, which trigger the reveal
05 Case Report	Reads a report written from their own run; prints or downloads it	Case #1 closed	Filing the report
The ending	Acknowledges the Suspension Notice, reads Agent Nano's note, then answers Nano's question	EXUN suspended; a digital jailbreak	A final choice: Case #2 pending, or off the record

05 Story design

Three acts

Act	Where	What changes
I. Recruitment	00 Intake, 01 Case Files	A codename, an ID card and a briefing; then the evidence, exactly as the official record tells it
II. Investigation	02 Energy Forensics, 03 Signal Decoder, 04 Machine Terminal	The numbers don't add up, the ID turns out to be a key, and the game's own code is missing its character

III. Fallout	The reveal, 05 Case Report, the ending	Nano's log confirms the discovery, the report is filed, EXUN is suspended, and the player chooses what happens next

How the story moves

- **Sealed folders.** Nothing opens before it is earned.
- **Lifting redactions.** Three passages start blacked out and lift as clearance grows: POI-1's "thick accent" after folder 02, "looking at bare wrist" after folder 03 (where it becomes new evidence), and Nano and Mega's "Maybe you computer guys could figure something out?" after folder 04.
- **The glitch bleed.** From folder 02 on, each cleared folder lets more of the arcade machine leak into the agency's paperwork, in three levels (see section 07).
- **An earned reveal.** After LOGS, DIFF and RENDER, the screen breaks up: signal interrupt. Only then does Agent Nano's log unseal, and a red-pen note appears on Transcript A: "He WASN'T a cosplayer!!!". Canon confirms the player's discovery instead of spoiling it.
- **An ending that turns.** Filing the report brings the Suspension Notice: on 27 October the Government suspends EXUN and calls the findings inconclusive. The player must acknowledge receipt, and from then on a suspension bar runs across every page.
- **A secret, and a choice.** Agent Nano's goodbye note slides out from behind the notice, in handwriting: the case was "just the first of a digital jailbreak", and the Government doesn't want the news getting out. It ends "What are you going to do now?", and the player answers: comply, or keep investigating off the record. The choice is final, and it shows on every page and in the report.

Three voices

Every line of story text is written in one of three voices, so the world feels consistent from the first screen to the last.

Voice	Where it speaks	What it sounds like (lines from the site)
The agency	Briefings, folder notes, the case report	"Read every document in the case files before you form a theory, and pin each detail that matters to the evidence board."
The Department of Electricity	Its reply to the player's referral	"The Department notes, with thanks, the expedience of the referral."
The machine	The arcade cabinet's green screen	"WE KNOW WHEN HE LEFT, AND HOW. NOBODY YET KNOWS WHY."

The timeline we never break

When	What happened	Source
23 October	A huge energy spike at Wenoti Arcade	Energy Usage Dossier
23 October, 22:31	Agents Nano and Mega meet POI-1 at the arcade machine	Transcript A
23 October	The machine begins “warbling” and “reverberating” after POI-1 leaves	Transcript A
24 October	The game source code is obtained: one character class is missing	Progress Assesment Log
25 October, 23:54	Agent Nano writes the log: the character escaped	Progress Assesment Log
27 October	EXUN is suspended by order of the Government of Nuxeland	Suspension Notice

Rules we set ourselves

- Never contradict canon: no new dates, names or facts that clash with the files.
- Never answer WHY the character escaped, or say where he went. That is Case #2.
- No hint before folder 04 that POI-1 came out of the game.
- Any document that would spoil the story stays sealed, or isn’t listed at all, until the player has earned it.

06 Puzzle design

Every puzzle is built from the material we were given, and the site works the answers out rather than storing them.

Puzzle	From the case files	What the player does	What happens underneath
Evidence board	Case Debrief, Transcript A, Asset Integrity Report (and later, the Suspension Notice)	Clicks details in the documents to pin them to a corkboard, then drags the cards, or moves them with the arrow keys	Card positions are saved as fractions of the board, so the cards stay put on any screen size. Filing needs all 7 key details.

Energy forensics	Energy Usage Dossier: 14 days of meter readings and reported usage	Runs the analysis, reads the chart (linear or logarithmic), and files a Discrepancy Referral naming the surge and the mismatch	Metered usage = today's reading − yesterday's. A modified z-score, 0.6745 × (x − median) ÷ MAD, flags any day above 3.5: 23 October scores 597. A mismatch is any day where reported and metered usage differ by more than 0.5 kWh: on 24 October, 503.2 kWh was reported but only 30.7 kWh metered.
Signal decoder	The Machine ID from the Case Debrief	Decodes MJWHKZLTNBUXE5A with six tools or auto-detect, then enters the key	Base64 (including URL-safe), Base32, hex, ASCII codes, Caesar shift and reverse, each working both ways. Auto-detect ranks every result by how much it reads like English: blueshirt scores 0.89, the next best 0.66.
Machine terminal	OriginalCode.txt and AlteredCode.txt	Presses LOGS, DIFF and RENDER on the arcade cabinet	A real line-by-line comparison finds 17 lines removed: the whole Character class (lines 4–20), which the obtained code still uses on lines 9 and 74. RENDER draws the 24 × 12 world from the code's own numbers: the player @ belongs at (12, 6), and isn't there.
Hints	All of the above	Opens up to three hints per folder, one at a time	Hint figures are filled in from the data, so they can never drift from the case.

WHY THIS MATTERS

Given different data, the same code finds different anomalies, and our tests prove it. Nothing is a hard-coded answer dressed up as a puzzle.

07 Experience and visual design

Two worlds meet on one desk: the agency's paperwork and the arcade machine. As the case goes on, the second leaks into the first.

	The agency	The machine
Surface	Off-white paper with grain and creases, overlapping folder tabs, a corkboard with red string, a manila in tray	A green CRT screen with scanlines, set in an arcade cabinet
Type	Roboto Mono, matching the case PDFs; Agent Nano's handwriting in Indie Flower	VT323, a CRT typeface
Colour	Exun blue, near-black ink, red stamps, marker yellow	Phosphor green and amber on near-black
Signature details	Worn rubber stamps, a paperclip, an ID card stamped Admitted, a pink carbon-copy referral form, ticker tape, a page torn from a notebook	Arcade buttons for LOGS, DIFF and RENDER, and a "signal interrupt" reveal

The glitch bleed

Level	Unlocked when	What happens to the agency's pages
1	Energy Forensics is cleared	Stamps and redaction bars misprint now and then
2	Signal Decoder is cleared	Headings tear, folder tabs flicker, and faint scanlines appear
3	Machine Terminal is cleared	A rolling bar and phosphor-green ghosting

Most of each animation is stillness: the glitch is a short moment near the end, so the pages stay readable.

What we refused

Anything that would make it look like a template or a marketing site: hero banners, purple gradients, frosted glass, emoji icons, rows of three cards and gradient text. Every screen had to look like it belonged in the case file.

08 Accessibility, built in

Accessibility was part of the design from the first screen, not an add-on at the end.

Need	What we built
Keyboard only	Everything works without a mouse, including moving evidence cards with the arrow keys, and a “skip to the open folder” link on the first Tab press
Screen readers	Every control has a name; pinning, filing, decoding and unlocking are announced; the chart has a text description
Small screens	The whole case plays on a 360 px phone with no sideways scrolling
Motion sensitivity	One switch turns the glitches off, and they stop by themselves when the device asks for less motion
Low vision	All text is checked against WCAG AA contrast
Sound	Off until the player turns it on

09 Technical architecture

A static site with no backend, built so that every part can be read, tested and explained by the three of us.

The stack

Layer	Choice	Why
Build	Vite	Fast and simple, and it outputs plain static files
Language	Plain JavaScript and CSS	No framework to hide behind: every line is ours to explain
Tests	Vitest	231 automated tests across 12 files
Hosting	Vercel	The live site, rebuilt from GitHub on every push
Checks	GitHub Actions	Runs the tests and the build on every push, so a broken change shows up straight away
Fonts	Bundled with the site	No outside font service

How the code is organised

Folder	What it holds

content/	canon.json (every document, word for word), story.json (our story text), hints.json, assets.json, team.json
data/energy.json	The dossier's table, checked row by row against the PDF
src/lib/	Pure logic with no page code: anomaly detection, decoders, the code diff, the game-world reader, the evidence rules, the report, the ending
src/state/	Progress, the unlock rules and saving
src/ui/	One file per screen, plus shared pieces: the top bar, the evidence board, the chart, the document renderer
tests/	231 tests covering the logic, the unlock rules, the report and the integrity of the canon text

Saving progress

- Progress is saved in the browser. If the browser blocks storage, the site still works for that visit.
- Saved data is rebuilt safely on every load: a hand-edited save can't skip folders, and the ending is checked step by step.
- Supervisor mode keeps its own separate save, so a judge looking around never touches a player's progress.

10 Key decisions

Decision	Why we made it	The trade-off we accepted
No framework, no backend	Every line can be explained, and there is nothing to host or secure	More hand-written code for screens and state
Logic kept apart from the screens	Every rule can be tested without a browser	More, smaller files to find your way around
Answers worked out, never stored	The puzzles stay honest and testable	Harder to write than hard-coded answers
All story text in one file	The team can write and edit the story without touching code	Lines are referred to by name in the code, not seen in place

Nano's log shown sealed; the Suspension Notice not listed at all	A sealed tab builds anticipation for the log; a tab for the notice would give the ending away	Two different sealing behaviours to explain
Addresses like #/files	Works on any static host with no server setup	Slightly less tidy web addresses

11 Risks we planned for

Risk	How we handled it
Judges have little time	Supervisor mode opens every folder, and a 3½-minute demo route covers the whole case
The browser blocks storage	The site falls back to a fresh start and keeps working for that visit
Someone edits their save to skip ahead	Saves are rebuilt on load: cleared folders only count in an unbroken chain, and the ending is checked step by step
Spoilers	Sealed documents, an unlisted notice, and no hint of the twist before folder 04
Drifting from canon	Tests compare our quotes, dates and figures with the canon text
Motion sensitivity	Reduced-motion support and a one-button glitch switch
A broken release	The tests and the build run on every push

12 How it meets the judging criteria

Criterion	What we built	Where to see it
Creativity	An in-world desk instead of a website: the corkboard and red string, the arcade cabinet, the glitch bleed, and Nano's torn notebook page	Every folder; the glitches grow from folder 02 on
Originality	Puzzles drawn from the real case files: the meter maths, the Base32 Machine ID, a real code comparison, and a world drawn from the code	Folders 02, 03 and 04
Progression of Storyline	Sealed folders, lifting redactions, growing glitches, an earned reveal, Nano's log, the suspension, Nano's note and a final choice	Folder 04 onwards, and the Case Report

Functionality of the website	Anomaly detection, six decoders, code comparison, saved progress, print and download, hints, supervisor mode, accessibility, and 231 tests	Everywhere; ?supervisor=1 opens it all

13 What comes next

Case #1 ends with EXUN suspended and Nano's warning that characters are escaping arcades all over the world. Canon still leaves the biggest question open: **why?**

Next	The idea
Case #2	Starts from Nano's question and the "digital jailbreak": video game characters disappearing from arcades worldwide
Carry the choice forward	A suspended agent waiting for orders and one working off the record begin Case #2 in different places
Our own art and sound	Slots are ready for the team's agency mark, ID photo, two signatures, the arcade marquee and three sounds
Reuse the engine	Much of Case #1 lives in content files, so a new case can reuse the desk, the evidence board, the hints, the report and the progress system

14 Team AFBBS

Team	Medhansh Sharma, Aryaveer Dalal, Mudit Shakya
Play the case	wattsonbuild.vercel.app
See every folder	wattsonbuild.vercel.app/?supervisor=1
Read the code	github.com/medhu0505/Wattson_Build

CASE #1: CLOSED · CASE #2: PENDING

We know when he left, and how. Nobody yet knows why.